

Module Delivery Roadmap — Integrated University Management System

Assam Agricultural University (AAU) · Per-module requirement collection, build & delivery effort

Prepared by: Pallab Handique | **For:** BSPL Management (cost-estimation input) | **Version:** 0.1 (Draft) | **Date:** 11 June 2026 | **Baseline:** Visit Closure Document v1.0 + HLD (8–9 June 2026 study)

How to read this document. This is an **effort roadmap**, not a price. It sizes every module in **person-days** across four work types — **Req** (requirement collection), **Des** (design), **Dev** (development), **Tst** (test/QA) — and orders them by delivery phase and dependency. Management applies day-rates, overheads, margin and contingency on top to produce the formal estimate.

Confidence basis. Numbers are **indicative, pre-BRD**. They assume the working assumptions in the Closure Document hold. Items flagged **Low** confidence depend directly on the open points in Closure §6 (fund-source flow, module reality, formula governance, scope boundaries) and on securing the incumbent vendor's legacy code (Closure §7.2). Resolving those can move effort up or down materially.

1. Roadmap Approach & Legend

- Unit:** one person-day = one developer/analyst working day. Effort is build effort only — it excludes day-rate, leave, ramp-up and management margin (management adds those).
- Work types per module:** **Req** = requirement collection & analysis · **Des** = solution/data-model design · **Dev** = development · **Tst** = test, QA & defect fix.
- Granularity:** each module is broken into sub-features; sub-feature lines carry the Dev estimate; Req/Des/Tst are summed at module level.
- Sequencing:** phases follow the HLD dependency order. Foundation (Phase 0) gates everything; masters & EFMS must exist before payroll/finance workflows run.

Build state: **REUSE** existing AAU system — integrate, light effort **ENHANCE** exists but extended/wrapped

BUILD new module

Confidence: High · Med · Low

2. Delivery Phase Map

Five delivery waves. Foundation first because every later workflow depends on masters, the file rail and audit. Active employees go live before pensioners / inactive records.

PHASE 0 Foundation RBAC, EFMS file rail, Masters & Rule Engine, Audit Ledger, Notifications ~134 pd	PHASE 1 HRMS + Establishment Employee master integration (PAN), transfer/promotion/appraisal, JSON reuse ~61 pd	PHASE 2 Payroll + Leave + GPF Salary engine, deductions, corrections, increments, bill; parallel run	PHASE 3 Finance + FOC + Bill Fund control, budget, disbursement chain, treasury posting ~113 pd	PHASE 4 Pension + Store/Asset + ESS + Analytics Pension lifecycle, store, asset, portal, claims, loans, warehouse
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~118 pd

~193 pd

3. Effort Summary (Roll-Up)

Module build effort only. Program-level effort (PM, migration, integration, UAT, deployment, training) is in \$6 and added separately.

Phase / Module	State	Req	Des	Dev	Tst	Total	Conf.
P0 · User Mgmt / RBAC	ENHANCE	3	4	10	4	21	Med
P0 · EFMS file rail	BUILD	5	6	18	6	35	Med
P0 · Masters & Rule Engine	BUILD	8	10	25	8	51	Low
P0 · Audit Ledger	BUILD	2	3	8	3	16	High
P0 · Notification service	BUILD	1	2	6	2	11	High
Phase 0 subtotal	Foundation	19	25	67	23	134	
P1 · HRMS integration	REUSE	5	5	14	5	29	Med
P1 · Establishment	BUILD	5	5	16	6	32	Med
Phase 1 subtotal	HRMS + Estab.	10	10	30	11	61	
P2 · Payroll	BUILD	12	12	35	14	73	Low
P2 · Leave	BUILD	3	3	10	4	20	High
P2 · GPF	BUILD	4	4	12	5	25	Med
Phase 2 subtotal	Payroll core	19	19	57	23	118	
P3 · Finance & Accounts	BUILD	8	8	24	9	49	Low
P3 · FOC / Budget	BUILD	7	7	18	7	39	Low
P3 · Bill Tracking	BUILD	4	4	12	5	25	Med
Phase 3 subtotal	Finance + Fund	19	19	54	21	113	
P4 · Pension	BUILD	6	6	18	7	37	Low
P4 · Store & Purchase	BUILD	5	5	16	6	32	Med
P4 · Asset & Estate	BUILD	4	4	12	5	25	Med
P4 · Employee Portal (ESS)	ENHANCE	3	4	12	5	24	Med
P4 · Claims	BUILD	3	3	9	4	19	Med
P4 · Loans & Advances	BUILD	4	4	12	5	25	Low
P4 · Reports / Warehouse / Analytics	BUILD	4	5	16	6	31	Med
Phase 4 subtotal	Remaining scope	29	31	95	38	193	
MODULE BUILD TOTAL	16 modules	96	104	303	116	619	

4. Per-Module Detail (Sub-Feature Breakdown)

Each card lists the module's sub-features with Dev days, its dependencies, the key inputs it needs from AAU (ties to Closure §7), and external interfaces. Req/Des/Tst totals repeat the roll-up.

Phase 0 — Foundation (cross-cutting; gates all later modules)

User Management / RBAC

ENHANCE

Req 3 · Des 4 · Dev 10 · Tst 4 · 21 pd

Sub-feature	Dev	Note
Generalise EFMS 6-user-type / 3-access-level model into system-wide RBAC	4	core
Role / permission admin UI	3	
SSO / single login, session, password policy	2	reuse existing logins
Centre / department scoping (users see own domain)	1	

Depends on: none (first build). **Inputs:** EFMS role/access-level definitions. **Feeds:** every module.

EFMS — E-File Movement

BUILD

Req 5 · Des 6 · Dev 18 · Tst 6 · 35 pd

Sub-feature	Dev	Note
File lifecycle state machine (Received ▶ Accessed ▶ Forwarded ▶ Processed)	6	core rail
Access-based routing (3 levels) + forwarding rules	4	
Full movement log + creator lifecycle view	3	
Reporting-officer notification on processing	2	uses Notif. svc
Reusable workflow API (consumed by Bill/Estab/Leave/Pension)	3	key reuse

Depends on: RBAC. **Inputs:** EFMS role/access definitions, current file-movement walkthrough. **Open point:** EFMS vs salary-bill workflow — one engine or two (Closure §6.4).

Masters & Rule Engine

BUILD

Req 8 · Des 10 · Dev 25 · Tst 8 · **51 pd** · Low conf.

Sub-feature	Dev	Note
Master CRUD: DDO, Designation, Grade, Fund Type, Budget Head, Discipline, Nature Type, Payscale	7	
Mapping layer (combine masters → valid "setting")	4	
Formula engine (author, store, map formula→head)	7	highest risk
Deduction-breakup master (LIC/GIS/SSS/IT/loans)	3	
Effective-dating + versioning (past runs recompute on then-rules)	4	audit-critical

Depends on: RBAC, Audit Ledger. **Inputs:** master-table list, 2–3 real formula definitions. **Open point:** formula governance — who authors, versioning, mid-year recompute (Closure \$6.3). Drives most downstream modules.

Audit Ledger

BUILD

Req 2 · Des 3 · Dev 8 · Tst 3 · **16 pd**

Sub-feature	Dev	Note
Append-only log (who/what/when/why/which sanction)	4	
Maker-checker hook reusable by all modules	2	
Audit query / AG-CAG / RTI export views	2	

Depends on: RBAC. **Feeds:** every write path. Government-grade auditability.

Notification Service

BUILD

Req 1 · Des 2 · Dev 6 · Tst 2 · **11 pd**

Sub-feature	Dev	Note
Event → notification dispatch (email + in-app)	4	PAN+email link
Templates + reporting-officer targeting	2	

Depends on: RBAC. **Feeds:** EFMS, Bill Tracking, Leave, alerts.

Phase 1 — HRMS Integration + Establishment

HRMS Integration

REUSE

Req 5 · Des 5 · Dev 14 · Tst 5 · 29 pd

Sub-feature	Dev	Note
Employee-master read adapter (PAN = key), no re-keying	5	source of truth
Consume JSON profiles (qualifications, employment, research) as-is	4	
Department ↔ faculty shared-ID transfer-history reuse	3	
PAN-based de-duplication (one person = one record)	2	

Depends on: Masters. **Inputs:** JSON schema dumps, HRMS API maturity (Closure §6 open). **Open point:** connector effort depends on existing API maturity.

Establishment

BUILD

Req 5 · Des 5 · Dev 16 · Tst 6 · 32 pd

Sub-feature	Dev	Note
Transfer workflow (form + EFMS + audit)	4	
Promotion / increment workflow	4	
Appraisal integration — 8 existing formats (APAR/CAS/MACP feed)	5	integrate, not rebuild
Retirement-due auto-listing (age varies by cadre) → Pension	3	

Depends on: HRMS integ, EFMS, Masters. **Inputs:** 8 appraisal formats, retirement-age rules by cadre. **Open point:** scope of 8 formats to integrate (Closure §6).

Phase 2 — Payroll + Leave + GPF (core revenue module)

Payroll Engine

BUILDReq 12 · Des 12 · Dev 35 · Tst 14 · **73 pd** · Low conf.

Sub-feature	Dev	Note
Gross calc (pay + allowances) from masters/formulas	6	
Deduction engine (IT monthly, LIC, GIS, SSS, loans, GPF)	7	
Net + per fund-type / budget-head tagging	4	feeds FOC
Attendance-linked pay (LWP / absence reduces pay)	3	
Batch corrections	4	
Formula-based increments	3	
Arrears (recompute on effective-dated rules)	4	audit-critical
Dynamic payslip + annual employee report + salary-difference/scrutiny report	4	

Depends on: Masters/Rule engine, HRMS, Leave (LWP), GPF, Audit. **Inputs:** real payslip, real treasury pay-bill, formula defs, deduction sheets, sample arrear case. **Open points:** payscale model designation-vs-grade-wise (HLD↔Closure contradiction), IT deduction-only vs full statutory (§6.4). **Excludes:** TA/DA (separate billing module).

Leave

BUILDReq 3 · Des 3 · Dev 10 · Tst 4 · **20 pd**

Sub-feature	Dev	Note
Apply → approve → balance update workflow	5	uses EFMS
LWP detection → Payroll pay impact	3	
Leave-type masters + balances	2	

Depends on: HRMS, EFMS, Masters. **Feeds:** Payroll, Establishment.

GPF

BUILDReq 4 · Des 4 · Dev 12 · Tst 5 · **25 pd**

Sub-feature	Dev	Note
Subscription (Payroll deduction link)	3	
Advances / withdrawals	4	
Ledger + annual statement	5	feeds Pension

Depends on: Payroll, Masters, Audit. **Inputs:** GPF rules, opening balances. **Feeds:** Pension.

Phase 3 — Finance + FOC + Bill Tracking (money convergence)

Finance & Accounts

BUILDReq 8 · Des 8 · Dev 24 · Tst 9 · **49 pd** · Low conf.

Sub-feature	Dev	Note
General ledger + chart of accounts	7	
Fund accounting (per fund type/head)	5	
Postings intake (Payroll, Fee, Store)	4	convergence point
Treasury interaction	4	external
Reconciliation (payroll-to-bill, fund-to-spend totals must agree)	4	audit evidence

Depends on: Payroll, FOC, Masters. **Open point:** salary treasury-drawn vs university-drawn → depth of treasury integration (Closure §6).

FOC / Budget — Fixation of Ceiling

BUILDReq 7 · Des 7 · Dev 18 · Tst 7 · **39 pd** · Low conf.

Sub-feature	Dev	Note
Fund-source intake (govt budget, ICAR/scheme grants, receipts, endowments)	4	
Fund ↔ head ↔ sector + designation ↔ fund mapping	5	
Ceiling fixation + dispersal breakup on receipt	5	
Ceiling-gating check at each spend (salary/store/other)	4	gates Bill chain

Depends on: Masters, Finance. **Inputs:** complete fund-source list, one project's funding-to-disbursal trail. **Open point:** actual fund-source tracing is the biggest study gap (Closure §6.1).

Bill Tracking

BUILDReq 4 · Des 4 · Dev 12 · Tst 5 · **25 pd**

Sub-feature	Dev	Note
Disbursement chain state machine (Accounts→DDO→Budget→Cashier→Treasury)	6	reuses EFMS
Per-hop FOC ceiling check	3	
Single live status / full-lifecycle view to originator	3	

Depends on: EFMS, FOC, Payroll (bill+PDF). **Inputs:** real treasury pay-bill, observed approval chain.

Phase 4 — Pension, Store/Asset, ESS, Claims, Loans, Analytics

Pension

BUILDReq 6 · Des 6 · Dev 18 · Tst 7 · **37 pd** · Low conf.

Sub-feature	Dev	Note
Retirement intake from Establishment auto-list	3	
Pension computation (service history + GPF)	7	
Pensioner records + lifecycle	5	
NPS-CRA / Jeevan Pramaan / AG office interfaces	3	external

Depends on: Establishment, GPF, Finance, history load. **Open point:** pension scheme cohorts OPS/NPS/UPS + integrated vs handoff (Closure §6.4).

Store & Purchase

BUILDReq 5 · Des 5 · Dev 16 · Tst 6 · **32 pd**

Sub-feature	Dev	Note
Requisition → purchase → receipt → issue workflow	8	currently offline
FOC fund draw + Finance (AP) posting	4	
Feeds Asset register	4	

Depends on: FOC, Finance, Asset. **Inputs:** current store/purchase process detail. **Note:** excluded from initial phase unless explicitly requested (Closure §5).

Asset & Estate

BUILDReq 4 · Des 4 · Dev 12 · Tst 5 · **25 pd**

Sub-feature	Dev	Note
Asset register + allocation	6	
Estate / quarters + license-fee deduction → Payroll	6	

Depends on: Store, Finance, Payroll (license-fee).

Employee Portal (ESS)

ENHANCEReq 3 · Des 4 · Dev 12 · Tst 5 · **24 pd**

Sub-feature	Dev	Note
Self-service: profile, payslip, leave, claims (logins exist)	6	read across modules
Document downloads (payslip/annual statement/GPF)	3	
Request submission into workflows	3	

Depends on: Payroll, Leave, GPF, Claims. **Reuse:** existing employee logins.

Claims

BUILD

Req 3 · Des 3 · Dev 9 · Tst 4 · 19 pd

Sub-feature	Dev	Note
Claim submission → approval workflow	5	uses EFMS
Payment via bill chain / reimbursement posting	4	

Depends on: EFMS, Finance, Bill Tracking. **Note:** TA/DA billed here, outside salary engine.

Loans & Advances

BUILD

Req 4 · Des 4 · Dev 12 · Tst 5 · 25 pd · Low conf.

Sub-feature	Dev	Note
Loan grant + schedule	5	
Recovery via Payroll deduction	4	
Balance / closure ledger	3	

Depends on: Payroll, Masters. **Inputs:** sample loan schedule. **Open point:** full lifecycle vs deduction-line only (Closure \$6.4) — scope swings effort.

Reports / Warehouse / Analytics

BUILD

Req 4 · Des 5 · Dev 16 · Tst 6 · 31 pd

Sub-feature	Dev	Note
Warehouse / archive (cumulative history, read-mostly)	6	
Central dashboards (activity across modules, real time)	5	portal vision
Payroll / fund / pension / HR analytics + AG-CAG/RTI exports	5	

Depends on: all transactional modules live.

5. Requirement Collection Plan (per module)

Method per module, run before its build phase opens. Each module's Req-days above assume this sequence completes.

Step	What happens	Output / gate
1. Sample study	Study the real artifacts from Closure §7 (payslip, treasury bill, formula defs, JSON schema, master tables, appraisal formats)	Annotated as-is process notes
2. Legacy code review	Reverse-engineer incumbent vendor's DB schema, stored procs, batch jobs, report SQL (Closure §7.2 — time-critical)	Extracted business-logic catalogue (salary rules, rounding, arrears, fixation)
3. SME workshops	Online clarification sessions by domain: HR → Payroll → Finance → IT, resolving Closure §6 open points	Decision log entries, signed
4. Module spec	Per-module requirement note + acceptance criteria; updates effort if scope shifts	Module spec → BRD section
5. Sign-off gate	Decision authority confirms before build starts	Scope locked for that module

Critical-path dependency. Step 2 (legacy vendor handover) is the single biggest schedule risk. The genuine salary/fund/fixation logic lives in incumbent code, not manuals. If handover slips, Payroll, Finance, FOC and Pension requirement collection cannot complete and their build dates move. Flag this to AAU for early formal instruction to the vendor.

Per-module input checklist (ties to Closure §7)

Module	Must-have inputs before build
Masters / Payroll	Real payslip, real treasury pay-bill, 2–3 formula defs, master-table list, deduction sheets (GIS/LIC/SSS), sample arrear case, payscale model decision
HRMS / Establishment	JSON schema dumps, 8 appraisal formats, retirement-age rules by cadre, HRMS API maturity
FOC / Finance	Complete fund-source list, one project's funding-to-disbursal paper trail, treasury-drawn vs university-drawn decision
GPF / Loans / Pension	GPF rules + opening balances, sample loan schedule, pension cohort (OPS/NPS/UPS) decision
All	Module usage-status sheet (live / partial / unused / offline) — confirms what is actually built

6. Program-Level Effort (added on top of module build)

Effort not attributable to a single module. Management adds these to the §3 module total.

Item	Basis	Person-days
Project / delivery management	~12% across programme duration	74
Data migration	Legacy → opening balances (live) + full history (warehouse); PAN de-dup, cleanse, load	30
External integrations	Treasury, PFMS, Banks (NEFT/RTGS), NPS-CRA, Income Tax/TRACES, AG/Pension, Jeevan Pramaan, ICAR/UGC, payment gateways	45
System / integration testing + UAT support	Cross-module, reconciliation, parallel-run validation	25
Deployment, go-live, parallel run	Per phase cutover	20
Training & documentation	SME/end-user training, admin + user manuals	15
Program-level subtotal		209
INDICATIVE TOTAL (module 619 + program 209) before margin & contingency		828

For management: apply day-rate, overhead and margin to the 828 pd. Recommend a **contingency band of +15% to +25%** weighted to the **Low-confidence** modules (Masters/Rule engine, Payroll, Finance, FOC, Pension, Loans), pending Closure \$6 resolution and legacy-code access.

7. Indicative Resource Profile by Phase

Role	P0	P1	P2	P3	P4
Project Manager	●	●	•	•	•
Business Analyst	•	•	•	•	•
Solution / Data Architect	•	●	●	●	○
Backend Developers (×2–3)	•	•	•	•	•
Frontend Developer	●	●	•	●	•
DBA / Migration Engineer	●	•	●	•	•
Integration Engineer	○	○	●	•	•
QA / Tester	●	•	•	•	•

● full engagement · ● part engagement · ○ minimal. Exact headcount and calendar duration follow from agreed day-rate and parallelism — management to set.

8. Delivery Milestones & Acceptance

Phase	Delivered	Exit / acceptance criteria
P0 Foundation	RBAC, EFMS rail, Masters/Rule engine, Audit, Notifications	A master setting drives a sample calc; a file moves end-to-end with full audit log
P1 HRMS + Estab.	Employee master integrated (PAN), transfer/promotion/appraisal, JSON reuse	Employee resolvable by PAN across modules; retirement auto-list works
P2 Payroll core	Salary engine, deductions, corrections, increments, bill+PDF, Leave, GPF	Parallel run matches existing payslip + treasury bill within tolerance
P3 Finance + Fund	FOC ceiling control, budget, bill-tracking chain, treasury posting	Payroll-to-bill & fund-to-spend reconciliation totals agree
P4 Remaining	Pension, Store, Asset, ESS, Claims, Loans, warehouse/analytics	Pension computed from history+GPF; dashboards live; RTI/AG export verified

9. Risks, Assumptions & Exclusions (effort-affecting)

Risks that move effort

- **Legacy logic lives in incumbent code** — handover delay blocks Payroll/Finance/FOC/Pension req-collection (Closure §7.2).
- **Payscale model unresolved** — Closure says pay-grade-wise, HLD says designation-wise (to confirm). Reshapes Masters + Payroll.
- **Module reality unknown** — paper-only modules may add or remove scope until usage-status sheet received.
- **Fund-source tracing is the stated study gap** — FOC effort can swing on actual flow (Closure §6.1).
- **External API maturity unknown** — treasury/PFMS/NPS integration depth variable.
- **Scope boundaries open** — IT (deduction vs full statutory), loans (lifecycle vs deduction), pension (integrated vs handoff), EFMS-vs-salary-bill engine count (Closure §6.4).

Assumptions (per Closure §5)

- HRMS data reasonably complete and usable as primary source; PAN unique per employee.
- AAU provides data, samples, SMEs and decision authority on time.
- Salary/fund/formula logic made available for reverse-engineering.

Excluded from this effort (unless later agreed)

- TA/DA inside salary engine (separate billing — see Claims).
- Offline Store/Inventory in initial phase unless explicitly requested.
- Modules confirmed unused — not rebuilt by default.
- Hardware, network/connectivity, third-party licences.

10. Next Steps

1. Management applies rate, overhead, margin and contingency to the §6 total → format cost estimate.
2. Collect Closure §7 inputs; issue formal vendor-handover instruction (time-critical).
3. Run domain SME workshops; close §6 open points into the decision log.

4. Convert this roadmap into the BRD with locked scope & phasing; revise effort where scope changed.